

ANSWERS

1.

- (a) rate of decay / activity / decay (of nucleus) is not affected by external factors / environment / surroundings B2
(if states specific factor(s), rather than giving general statement above, then give 2 marks for two stated factors, but 1 mark only if one factor stated)
- (b) (i) gamma / γ B1
 (ii) alpha / α B1
 (iii) gamma / γ B1
 (iv) beta / β B1

2.

- (a) deviation shown correctly B1
- (b) smaller deviation (not zero deviation) M1
 acceptable path wrt position of N A1
- (c) the nucleus is (very) small M1
 in comparison to the atom A1
(special case: 'atom is mostly empty space' scores 1 mark)
- (d) deviation depends on charge on the nucleus / N / electrostatic repulsion B1
 same charge so no change in deviation B1

3.

- (a) *either* forms of same element
or atoms / nuclei with same number of protons M1
 atoms / nuclei contain different numbers of neutrons A1
(use of 'element' rather than atoms / nuclei scores max 1 mark)
- (b) (i) decay is not affected by environmental factors B1
(allow two named factors)
- (ii) *either* time of decay (of a nucleus) cannot be predicted
or nucleus has constant probability in a given time B1
- (c) $^{185}_{75}\text{Re}$ B1
either $^0_{-1}\text{e}$ *or* $^0_{-1}\beta$ B1

- 4.
- (a) nuclei/atoms with same proton number/atomic number B1
 nuclei/atoms contain different numbers of neutrons/different atomic mass B1
- (b) (i) 92 A1
 (ii) 146 A1
- (c) (i) mass = $238 \times 1.66 \times 10^{-27}$ C1
 = 3.95×10^{-25} kg A1
- (ii) volume = $\frac{4}{3} \pi \times (8.9 \times 10^{-15})^3$ (= 2.95×10^{-42}) C1
 density = $(3.95 \times 10^{-25}) / (2.95 \times 10^{-42})$
 = 1.3×10^{17} kg m⁻³ A1
- (d) nucleus contains most of mass of atom B1
 either nuclear diameter/volume very much less than that of atom
 or atom is mostly (empty) space B1
- 5.
- (a) (i) *either* helium nucleus
 or contains 2 protons and 2 neutrons B1
- (ii) e.g. range is a few cm in air/sheet of thin paper
 speed up to 0.1 c
 causes dense ionisation in air
 positively charged or deflected in magnetic or electric fields
 (any two, 1 each to max 2) B2
- (b) (i) ${}^4_2\alpha$ B1
 either 1_1p or 1_1H B1
- (ii) 1 initially, α -particle must have some kinetic energy B1
- (ii) 2 1.1 MeV = $1.1 \times 1.6 \times 10^{-13} = 1.76 \times 10^{-13}$ J C1
 $E_K = \frac{1}{2}mv^2$ C1
 $1.76 \times 10^{-13} = \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2$ C1
 $v = 7.3 \times 10^6$ m s⁻¹ A1
 use of 1.67×10^{-27} kg for mass is a maximum of 3/4

- 6.
- (a) *either* different forms of same element
or nuclei have same number of protons
different numbers of neutrons (in the nucleus) M1
A1
- (b) (i) proton number conserved B1
nucleon number conserved B1
mass-energy conserved B1
- (ii) 1. $Z = 36$ A1
2. $x = 3$ A1
- 7.
- (a) (i) most α -particles were deviated through small angles B2
(*allow 1 mark for 'straight through' / undeviated*)
- (ii) small fraction of α -particles deviated through large angles M'
greater than 90° (*allow rebound back*) A1
- (b) e.g. β -particles have a range of energies
 β -particles deviated by (orbital) electrons
 β -particle has (very) small mass
(*any two sensible suggestions, 1 each, max 2*) B2
- Do not allow β -particles have negative charge or β -particles have high speed
- 8.
- (a) nuclei with the same number of protons B1
and a different number of neutrons B1
- (b) (i) (mass + energy) (taken together) is conserved (B1)
momentum is conserved (B1)
one point required max. 1 B1
- (ii) $a = 1$ and $b = 0$ B1
 $x = 56$ B1
 $y = 92$ B1
- (c) proton number = 90 B1
nucleon number = 235 B1

9.

- (a) (i) the half life / count rate / rate of decay / activity is the same no matter what external factors / environmental factors **or** two named factors such as temperature and pressure changes are applied B1
- (ii) the observations of the count rate / count rate / rate of decay / activity / radioactivity during decay shows variations / fluctuations B1

(b)

property	α -particle	β -particle	γ -radiation
charge	(+)2e	-e	0
mass	4u	$9.11 \times 10^{-31} \text{ kg}$	0
speed	0.01 to 0.1 c	up to 0.99 c	c

one mark for each correct line

B3

- (c) collision with molecules B1
causes ionisation (of the molecule) / electron is removed B1

10.

- (a) (i) 2 protons and 2 neutrons B1
- (ii) e.g. positively charged 2e
mass 4u
constant energy
absorbed by thin paper *or* few cm of air (3 cm $\rightarrow$ 8 cm)
(*not low penetration*)
highly ionizing
deflected in electric/magnetic fields
(*One mark for each property, max 2*) B2
- (b) mass-energy is conserved B1
difference in mass 'changed' into a form of energy B1
energy in the form of kinetic energy of the products / γ -radiation
photons / e.m. radiation B1

11.

- (a) W = 1 and X = 0 A1
Y = 2 A1
Z = 55 A1
- (b) explanation in terms of mass – energy conservation B1
energy released as gamma or photons or kinetic energy of products or
em radiation B1

12.

- (a) thin paper reduces count rate hence α B1
 addition of 1 cm of aluminium causes little more count rate reduction hence only other radiation is γ B1
- (b) magnetic field perpendicular to direction of radiation B1
 look for a count rate in expected direction / area if there were negatively charged radiation present. If no count rate recorded then β not present. B1

13.

- (a) 92 protons in the nucleus and 92 electrons around nucleus B1
 143 neutrons (in the nucleus) B1
- (b) (i) α -particle travels short distance in air B1
 (ii) very small proportion in backwards direction / large angles B1
 majority pass through with no /small deflections B1
 either most of mass is in very small volume (nucleus) and is charged or most of atom empty space B1
- (c) $I = Q / t$ C1
 $n/t = (1.5 \times 10^{-12}) / (2 \times 1.6 \times 10^{-19})$ C1
 $n/t = 4.7 \times 10^6 \text{ s}^{-1}$ A1

14.

- (a) ${}^3_2\text{He} + {}^3_2\text{He} \rightarrow {}^4_2\text{He} + 2 {}^1_1\text{p} + Q$
 A numbers correct (4 and 1) B1
 Z numbers correct (2 and 1) B1
- (b) both nuclei have 2 protons B1
 the two isotopes have 1 neutron and two neutrons B1
 [allow 1 for 'same number of protons but different number of neutrons']
- (c) proton number and neutron number B1
 energy – mass B1
 momentum B1
- (d) (i) γ radiation B1
 (ii) product(s) must have kinetic energy B1
- (e) $13.8 \text{ MeV} = 13.8 \times 1.6 \times 10^{-19} \times 10^6 (= 2.208 \times 10^{-12})$ C1
 $60 = n \times 13.8 \times 1.6 \times 10^{-13}$
 $n = 2.7(2) \times 10^{13} \text{ s}^{-1}$ A1

15.

- (a) (i) electron B1
 (ii) any **two**:
 can be deflected by electric and magnetic fields or negatively charged /
 absorbed by few (1 – 4) mm of aluminum / 0.5 to 2 m or metres for range in air /
 speed up to 0.99c / range of speeds / energies B2
 (iii) decay occurs and cannot be affected by external / environmental factors
 or two stated factors such as chemical / pressure / temperature / humidity B1
- (b) 3 and 0 for superscript numbers B1
 2 and –1 for subscript numbers B1
- (c) energy = $5.7 \times 10^3 \times 1.6 \times 10^{-19}$ (= 9.12×10^{-16} J) C1

$$v^2 = \frac{2 \times 9.12 \times 10^{-16}}{9.11 \times 10^{-31}}$$
 C1

$$v = 4.5 \times 10^7 \text{ ms}^{-1}$$
 A1
- (d) both have 1 proton and 1 electron B1
 1 neutron in hydrogen-2 and 2 neutrons in hydrogen-3 B1
 (special case: for one mark 'same number of protons / atomic number
 different number of neutrons')

16.

- (a) the majority/most went straight through
 or were deviated by small angles B1
 a very small proportion/a few were deviated by large angles B1
 small angles described as $< 10^\circ$ and large angles described as $> 90^\circ$ B1
- (b) most of the atom is empty space/nucleus very small compared with atom B1
 mass and charge concentrated in (very small) nucleus B1
 correct links made with statements in (a) B1

17.

- (a) (i) $W = 206$ and $X = 82$
 $Y = 4$ and $Z = 2$ A1
- (ii) mass-energy is conserved A1
 mass on rhs is less because energy is released B1
- (b) not affected by external conditions/factors/environment B1
 or two examples temperature and pressure B1

18.

- (a) (i) the direction of the fields is the same OR fields are uniform OR constant electric field strength OR $E = V / d$ with symbols explained B1
- (ii) reduce p.d. across plates B1
increase separation of plates B1
- (iii) α opposite charge to β (as deflection in opposite direction) B1
 β has a range of velocities OR energies (as different deflections) and
 α all have same velocity OR energy (as constant deflection) B1
 α are more massive (as deflection is less for greater field strength) B1
- (b) $W = 234$ and $X = 90$ B1
 $Y = 4$ and $Z = 2$ B1
- (c) $A = 32$ and $B = 16$ and $C = 0$ and $D = -1$ B1

19.

- (a) α : helium nucleus
 β : electron
 γ : electromagnetic radiation / wave / ray or photon
three correct 2/2, two correct 1/2 B2
- (b) (i) atomic number / proton number / $Z - 2$, nucleon / mass number / $A - 4$ B1
- (ii) atomic number / proton number / $Z + 1$ B1
nucleon / mass number / A no change
- (iii) no change in proton or mass number or "no change" B1

20.

- (a) (i) A: 206, nucleon(s) or neutron(s) and proton(s) }
B: 82, proton(s) } all correct A1
- (ii) kinetic / E_K / KE B1
- (b) energy = $5.3 \times 1.6 \times 10^{-13}$ (J) [= 8.48×10^{-3} (J)] C1
power = $(7.1 \times 10^{18} \times 5.3 \times 1.6 \times 10^{-13}) / (3600 \times 24)$ = 70 (69.7) W

21.

(a) 92 protons and 143 neutrons

B1

(b)

	value
a	1
b	0
c	141
d	55

(a and b both required)

B1

B1

B1

(c) kinetic energy (of products) or gamma/ γ (radiation or photon)

B1

(d) (total) mass on left-hand side/reactants is greater than (total) mass on right-hand side/products

M1

difference in mass is (converted to) energy

A1

22.

(a) (i) (rate of decay) not affected by any external factors **or** changes in temperature and pressure etc.

B1

(ii) two protons and two neutrons

B1

(b) (i) (total) mass before decay/on left-hand side is greater than (total) mass on right-hand side/after the decay

M'

the difference in mass is released as kinetic energy of the products

A1

(may also be some γ radiation) (to conserve mass-energy)

(ii) $(6.2 \times 10^6 \times 1.6 \times 10^{-19} =) 9.9(2) \times 10^{-13} \text{ J}$

A1

23.

(a) ${}^1_1\text{p}$

B1

${}^0_{-1}\beta^-$ and ${}^0_0\bar{\nu}$

B1

(b) an (electron) antineutrino

B1

(c) lepton(s)

B1

(d) (i) down, down, up/ddu

B1

(ii) a down/d (quark) changes to an up/u (quark) **or** ddu $\rightarrow$ uud

B1

24.

- (a) hadron: neutron/proton
and
lepton: electron/(electron) neutrino
(allow other correct particles) B1
- (b) (i) proton: up up down or uud B1
(ii) neutron: up down down or udd B1
- (c) (i) neutron $\rightarrow$ proton + electron + (electron) antineutrino B1
(ii) up down down (quarks) change to up up down (quarks)
or
down (quark) changes to up (quark) B1

25.

- (a) both electron and neutrino: lepton(s) B1
both neutron and proton: hadron(s)/baryon(s) B1
- (b) (i) ${}^1_1\text{p} \rightarrow {}^1_0\text{n} + {}^0_1\beta + {}^0_0\nu$
correct symbols for particles M1
correct numerical values (allow no values on neutrino) A1
(ii) up up down or uud $\rightarrow$ up down down or udd B1
(iii) weak (nuclear) B1

26.

- (a) α -particle is 2 protons and 2 neutrons; β^- -particle is positive electron/positron
 α -particle has charge +2e; β^+ -particle has +e charge
 α -particle has mass 4u; β -particle has mass (1/2000)u
 α -particle made up of hadrons; β^+ -particle a lepton
any three B3
- (b) ${}^1_1\text{p} \rightarrow {}^1_0\text{n} + {}^0_1\beta + {}^0_0\nu$
all terms correct M1
all numerical values correct (ignore missing values on ν) A1
- (c) (i) 1. proton: up, up, down / uud B1
2. neutron: up, down, down / udd B1
(ii) up quark has charge +2/3 (e) and down quark has charge -1/3 (e)
total is +1(e) B1

27.

- (a) hadron not a fundamental particle/lepton is fundamental particle
 or
 hadron made of quarks/lepton not made of quarks
 or
 strong force/interaction acts on hadrons/does not act on leptons B1
- (b) (i) proton: up, up, down / uud B1
 neutron: up, down, down / udd B1
- (ii) composition: $2(\text{uud}) + 2(\text{udd})$
 $= 6 \text{ up, } 6 \text{ down} / 6\text{u, } 6\text{d}$ B1
- (c) (i) most of the atom is empty space
 or
 the nucleus (volume) is (very) small compared to the atom B1
- (ii) nucleus is (positively) charged B1
 the mass is concentrated in (very small) nucleus/small region/small volume/small core
 or
 the majority of mass in (very small) nucleus/small region/small volume/small core B1

28.

- (a) hadron not a fundamental particle/lepton is fundamental particle
 or hadron made of quarks/lepton not made of quarks
 or strong force/interaction acts on hadrons/does not act on leptons B1
- (b) (i) ${}^0_1\text{e}^{(+)}$ or ${}^0_1\beta^{(+)}$ B1
 ${}^0_0\nu_{(\text{e})}$ B1
- (ii) weak (nuclear force / interaction) B1
- (iii) • mass-energy
 • momentum
 • proton number
 • nucleon number
 • charge
Any three of the above quantities, 1 mark each B3
- (c) (quark structure of proton is) up, up, down or uud B1
 up/u (quark charge) is $(+) \frac{2}{3}(e)$, down/d (quark charge) is $- \frac{1}{3}(e)$ C1
 $\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = (+)e$ A1

29.

- (a) (i) (proton is uud so) $(2/3)e + (2/3)e - (1/3)e = e$
 (a) (ii) (neutron is udd so) $(2/3)e - (1/3)e - (1/3)e = 0$

B1

B1

B1

(b)(i)

	β^-	β^+
nucleon number	90	64
proton number	39	28

all correct

- (ii) weak (nuclear force/interaction)

B1

- (iii) β^- decay: electron and (electron) antineutrino

B1

β^+ decay: positron and (electron) neutrino

30.

Question	Answer	Marks
8(a)	β^- emission: neutron changes to proton (+ beta ⁻ /electron) and β^+ emission: proton changes to neutron (+ beta ⁺ /positron)	B1
	β^- emission: (electron) antineutrino also emitted and β^+ emission: (electron) neutrino also emitted	B1
8(b)	proton: up up down (and zero strange) neutron: up down down (and zero strange)	B1

31.

7(a)	lepton(s)	B1
7(b)	protons: 7 and neutrons: 6	A1
7(c)	$E = \frac{1}{2}mv^2$	C1
	$= 0.80 \times 10^6 \times 1.60 \times 10^{-19}$	C1
	$= 1.28 \times 10^{-13} \text{ (J)}$	A1
	$v^2 = 2 \times 1.28 \times 10^{-13} / 2.2 \times 10^{-26}$ $v = 3.4 \times 10^6 \text{ m s}^{-1}$	
7(d)	an (electron) neutrino/ $\nu_{(e)}$ is also produced (and this has energy)	B1

32.

7(a)	nucleons = 23	B1
	neutrons = 11	B1
7(b)	similarity: same (rest) mass or equal (magnitude of) charge	B1
	difference: opposite (sign of) charge or one is matter and one is antimatter or one is an electron and one is an antielectron	B1

33.

7(a)	arrow pointing vertically down the page	B1
7(b)	$E = \frac{1}{2}mv^2$	C1
	$E = 460 \times 1.60 \times 10^{-19} (= 7.36 \times 10^{-17} \text{ (J)})$	C1
	$v = [(2 \times 460 \times 1.60 \times 10^{-19}) / (9.11 \times 10^{-31})]^{1/2}$ $= 1.3 \times 10^7 \text{ ms}^{-1}$	A1
7(c)	β^- particles have range of/different/various speeds/velocities/momenta/energies	M1
	so they follow different paths	A1

34.

7(a)	circle(s) drawn only around β^- and $\bar{\nu}$ symbols	B1
7(b)	(electron) <u>antineutrino</u>	B1
7(c)	kinetic (energy)	B1
7(d)	Y has one more proton (and one less neutron)/X has one less proton (and one more neutron) or Y has more protons (and fewer neutrons)/X has fewer protons (and more neutrons) or a neutron changes to a proton or the number of protons increases	M1
	(so) not isotopes	A1
7(e)	up down down changes to up up down or udd $\rightarrow$ uud or down changes to up or d $\rightarrow$ u	B1

35.

7(a)(i)	Q plotted at (82, 210)	A1
7(a)(ii)	R plotted at (83, 210)	A1
7(b)	lepton(s)	B1
7(c)	up down down changes to up up down or udd $\rightarrow$ uud or down changes to up or d $\rightarrow$ u	B1

36.

5(a)	region (of space) where a force acts on a (stationary) charge	B1
5(b)	$E = F/Q$	B1
	$F = ma$ and (so) $a = \frac{Eq}{m}$	A1
5(c)(i)	protons = 96	A1
	neutrons = 148	A1
5(c)(ii)	mass-energy is conserved/mass change is 'seen' as energy	B1
	energy released as gamma (radiation)/KE of α /KE of Pu	B1
5(c)(iii)	ratio = $\frac{2}{4} \times \frac{240}{94}$ or ratio = $\frac{2 \times 1.60 \times 10^{-19}}{4 \times 1.66 \times 10^{-27}} \times \frac{240 \times 1.66 \times 10^{-27}}{94 \times 1.60 \times 10^{-19}}$	C1
	ratio = 1.3	A1

37.

8(a)	antineutrino and positron both underlined (and no other particles)	B1
8(b)(i)	nucleon number = 27	A1
	proton number = 13	A1
8(b)(ii)	weak (nuclear force/interaction)	B1
8(b)(iii)	an (electron) antineutrino / $\bar{\nu}_{(e)}$ is produced (and this has energy)	B1
	X has kinetic energy	B1

38.

5(a)	$E = F / Q$	M1
	$F = ma$ and (so) $q / m = a / E$	A1
5(b)	$m = (4 \times 1.60 \times 10^{-19} \times 3.5 \times 10^4) / 1.5 \times 10^{12} (= 1.49 \times 10^{-26} \text{ kg})$	B1
	$= 1.49 \times 10^{-26} / 1.66 \times 10^{-27} = 9.0 \text{ (u)}$	A1
5(c)	protons: 4 and neutrons: 5	A1
5(d)(i)	nuclei have the same charge and so same (magnitudes of) force	B1
5(d)(ii)	nuclei have different masses and same force and so different (magnitudes of) acceleration	B1

39.

7(a)	nucleus is charged	B1
	the mass is <u>concentrated</u> in (very small) nucleus	B1
	or the <u>majority</u> of the mass is in (very small) nucleus	
7(b)(i)	$-(1 / 3)e$	B1
7(b)(ii)	$2q - (1 / 3)e = e$ so $q = (2 / 3)e$	M1
	up / u (quarks) (allow charm or top quarks)	A1

40.

6(a)	path/direction in which a (free) <u>positive</u> charge will move	B1
6(b)	(lines) closer together in Y/further apart in X	B1
6(c)(i)	$a = Eq / m$	C1
	or $F = Eq$ and $F = ma$	
	ratio = $(1e / 0.15 \text{ u}) \times (4 \text{ u} / 2e)$ or $1 / 0.15 \times 4 / 2$	C1
	ratio = 13	A1
6(c)(ii)	down quark charge is $-(1 / 3)e$	C1
	$-(1 / 3)e + q = -1e$ so $q = -(2 / 3)e$	A1
	$-(2 / 3)e$ is anti-up / $\bar{u}$ (quark) (allow charm or top antiquark)	B1

41.

7(a)	beta/ β	B1
7(b)	alpha/ α	B1
7(c)	gamma/ γ	B1
7(d)	beta/ β	B1

42.

7(a)(i)	proton number = 17 and nucleon number = 35	A1
7(a)(ii)	(electron) neutrino	B1
7(b)	d/down (quark charge) is $-\frac{1}{3}(e)$ or <u>two</u> d/down (quark charges) is $-\frac{2}{3}(e)$ or s/strange (quark charge) is $-\frac{1}{3}(e)$	C1
	charge = $-\frac{1}{3}(e) - \frac{1}{3}(e) - \frac{1}{3}(e)$ $= -1(e)$	A1

43.

7(a)	number of protons = 92	A1
	number of neutrons = 142	A1
7(b)	$5.6 \text{ MeV} = 5.6 \times 1.60 \times 10^{-19} \times 10^6 (= 8.96 \times 10^{-13} \text{ J})$	C1
	number = $0.15 / (5.6 \times 1.60 \times 10^{-13})$ $= 1.7 \times 10^{11}$	A1
	or	
	$0.15 \text{ W} = 0.15 / (1.60 \times 10^{-19} \times 10^6) (= 9.38 \times 10^{11} \text{ MeV s}^{-1})$	(C1)
	number = $9.38 \times 10^{11} / 5.6$ $= 1.7 \times 10^{11}$	(A1)

44.

7(a)	number of protons = 95	A1
	number of neutrons = 146	A1
7(b)	Np/neptunium (nucleus) has <u>kinetic</u> energy or gamma/ γ -radiation produced	B1
7(c)(i)	$I = NQ / t$	C1
	$I = (6.9 \times 10^{11} \times 2 \times 1.60 \times 10^{-19}) / 30$ $= 7.4 \times 10^{-9} \text{ A}$	A1
7(c)(ii)	$P = (6.9 \times 10^{11} \times 5.5 \times 10^6 \times 1.60 \times 10^{-19}) / 30$	C1
	$= 0.020 \text{ W}$	A1

45.

7(a)(i)	most of the atom is empty space or the nucleus (volume) is very small <u>compared to the atom</u>	B1
7(a)(ii)	the nucleus is charged	B1
	the mass is <u>concentrated</u> in nucleus / small region / small volume / small core or the <u>majority</u> of the mass is in nucleus / small region / small volume / small core	B1
7(b)(i)	proton number = 84	A1
	nucleon number = 214	A1
7(b)(ii)	up down down changes to up up down / udd → uud or down changes to up / d → u	B1

46.

6(a)	the nucleus is charged	B1
	the majority of the mass (of atom) is in the nucleus	B1
6(b)	made up of quarks (so) not a fundamental particle	B1
6(c)	(Q =) $6.9 \times 10^{-9} \times 60$	C1
	number = $(6.9 \times 10^{-9} \times 60) / (2 \times 1.60 \times 10^{-19})$	C1
	= 1.3×10^{12}	A1

47.

6(a)(i)	up up down	B1
6(a)(ii)	up down down	B1
6(a)(iii)	(alpha-particle is) 2 protons and 2 neutrons	C1
	6 up, 6 down	A1
6(b)(i)	most of an <u>atom</u> is empty space or the nucleus (volume) is (very) small <u>compared with the atom</u>	B1
6(b)(ii)	the nucleus is charged	B1
	the majority of the mass of atom is in the nucleus	B1

48.

6(a)	α -particle mass given as 4u	B1												
	α -particle charge given as (+)2e	B1												
	both β -particles mass given as 0.0005 u	B1												
	β^+ charge given as (+)e and β^- charge given as -e (Completed table: <table border="1"> <tr> <td></td><td>mass / u</td><td>charge / e</td></tr> <tr> <td>α</td><td>4</td><td>(+)2</td></tr> <tr> <td>β^+</td><td>0.0005</td><td>(+)1</td></tr> <tr> <td>β^-</td><td>0.0005</td><td>-1</td></tr> </table>)		mass / u	charge / e	α	4	(+)2	β^+	0.0005	(+)1	β^-	0.0005	-1	B1
	mass / u	charge / e												
α	4	(+)2												
β^+	0.0005	(+)1												
β^-	0.0005	-1												
6(b)(i)	neutron decays into proton and an electron / β^- particle	B1												

6(b)(ii)	down to up	B1
6(b)(iii)	(electron) antineutrino(s) emitted	B1
	energy (released in decay)/momentum shared between antineutrino and β^- particle	B1

49.

7(a)	(total) momentum before (decay) is zero or P has zero momentum	B1
	(total momentum after decay must be zero so) α -particle and Q have momenta in opposite directions (and therefore velocities are in opposite directions)	B1
7(b)	$p = 239 \text{ (u)} \times v$ or $4 \text{ (u)} \times 1.6 \times 10^7$	C1
	$239 \text{ (u)} \times v = 4 \text{ (u)} \times 1.6 \times 10^7$	A1
	$v = 2.7 \times 10^5 \text{ m s}^{-1}$	
7(c)	$E_{(K)} = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times (1.6 \times 10^7)^2$	C1
	$= 8.5 \times 10^{-13} \text{ (J)}$	A1
	$= 8.5 \times 10^{-13} / 1.60 \times 10^{-13} \text{ (MeV)}$	
	$= 5.3 \text{ MeV}$	
7(d)(i)	1. R plotted at (95,147)	B1
	2. S plotted at (96,146)	B1
7(d)(ii)	(electron) <u>anti</u> neutrino	B1

50.

7(a)(i)	(electron) neutrino	B1
7(a)(ii)	nucleon number = 22	A1
	proton number = 10	A1
7(a)(iii)	up up down changes to up down down or up changes to down	B1
7(b)(i)	charge = $- \frac{2}{3} e$	A1
7(b)(ii)	antiup / anticharm / antitop	B1

51.

7(a)(i)	change in $A = 0$	A1
	change in $Z = (+)1$	A1
7(a)(ii)	(electron) <u>anti</u> neutrino	B1
7(b)(i)	up / u (charge) = $(+) \frac{2}{3} e$ or antidown / $\bar{d} = (+) \frac{1}{3} e$ or $(q) = \frac{2}{3} e + \frac{1}{3} e$	M1
	$q = (+)1e$	A1
7(b)(ii)	hadron(s)	B1
	meson(s)	B1

52.

7(a)(i)	P = 0 and Q = 137	A1
	R = -1 and S = 56	A1
7(a)(ii)	lepton(s)	B1
7(b)(i)	(charge of ddd / Y =) $-\frac{1}{3}(e) - \frac{1}{3}(e) - \frac{1}{3}(e) = -1(e)$	B1
	(charge of $\bar{u}d$ / Z =) $-\frac{1}{3}(e) - \frac{2}{3}(e) = -1(e)$	B1
7(b)(ii)	meson: Z / $\bar{u}d$ because consists of a quark and an antiquark	B1
	baryon: Y / ddd because consists of three quarks	B1

53.

7(a)(i)	${}^{18}_9\text{F} \rightarrow {}^{18}_8\text{O} + {}^0_{(+1)}\beta^{(+)} + {}^{(0)}_{(0)}\nu(e)$ v or neutrino (B1) ${}^0_{(+1)}\beta^{(+)}$ (B1) ${}^{18}_8\text{O}$ (B1)	B3
7(a)(ii)	up quark to down quark	B1
7(b)(i)	must be three (anti)quarks as largest (negative) quark charge is $(-)\frac{2}{3}(e)$ or mesons can only have a charge of 0 or $\pm 1(e)$	M1
	(so hadron is) a baryon	A1
7(b)(ii)	any combination of three from: antiup (quark) / up antiquark and/or anticharm (quark) / charm antiquark and/or antitop (quark) / top antiquark	B1

54.

6(a)	particle with no internal structure / particle which cannot be broken down into anything smaller	A1
6(b)	charges: $u = (+)\frac{2}{3}(e)$ or $d = -\frac{1}{3}(e)$ or $s = -\frac{1}{3}(e)$	C1
	$(+)\frac{2}{3}(e) - \frac{1}{3}(e) - \frac{1}{3}(e) = 0(e)$	A1
6(c)(i)	- same/equal mass - same/equal (magnitude of) charge - both fundamental (particles) - opposite (sign of) charge - one is matter and the other is antimatter Any two points, 1 mark each.	B2
6(c)(ii)	neutron/baryon consists of three quarks	B1
	pion/meson consists of one quark and one antiquark	B1

55.

7(a)	92 protons and 146 neutrons (in nucleus)	B1
	92 (orbital) electrons	B1
7(b)	charge = $2e$ (= $2 \times 1.60 \times 10^{-19}$ C)	C1
	mass = $4u$ (= $4 \times 1.66 \times 10^{-27}$ kg)	C1
	ratio = $(2 \times 1.60 \times 10^{-19}) / (4 \times 1.66 \times 10^{-27})$ = 4.8×10^7 C kg ⁻¹	A1
7(c)(i)	up down down / udd	B1
7(c)(ii)	up up up / uuu	B1

56.

6(a)	${}^{14}_7\text{X}$	B1
	${}^0_{-1}\text{e}^-$	B1
6(b)(i)	$d \rightarrow u + e^- + \bar{\nu}$ or $udd \rightarrow uud + e^- + \bar{\nu}$	B1
6(b)(ii)	$-1/3 (e) = +2/3 (e) - 1(e) (+0)$ or $2/3 (e) - 1/3 (e) - 1/3 (e) = 2/3 (e) + 2/3 (e) - 1/3 (e) - 1 (e) (+0)$	B1
6(c)(i)	electrons / β -particles (emitted from the nucleus) have a (continuous) range of / different (kinetic) energies	B1
6(c)(ii)	the (emitted) neutrinos take varying amounts of the (same total) energy (released in the decay)	B1

57.

7(a)(i)	X has same number of protons as Y (and so) charge of X is the same as the charge of Y	B1
7(a)(ii)	X has (one) more proton (than Z)	M1
	(so) X has greater charge (than Z)	A1
7(b)(i)	meson(s)	B1
7(b)(ii)	one quark and one antiquark	B1

58.

8(a)(i)	Y and Z have equal numbers of protons and (so) they have the same charge	B1
8(a)(ii)	Y has (two) fewer protons (than X)	M1
	(so) Y has less charge (than X)	A1
8(b)	(total) momentum before decay is zero or X has zero / no momentum	B1
	(total momentum after decay must be zero so) Y must have equal (and opposite) momentum to α -particle (so cannot be stationary / must have speed/velocity)	B1

59.

8(a)(i)	number of protons: equal/same	B1
	number of neutrons: unequal/different	B1
8(a)(ii)	down (quark) changes to up (quark) or up down down (quarks) change to up up down (quarks)	B1
8(a)(iii)	(electron) antineutrino	B1
8(b)	charm (quark charge) is $(+)2/3(e)$ or 2 charm (quark charges) is $(+)4/3(e)$ or bottom (quark charge) is $-1/3(e)$	C1
	charge = $+2/3(e) + 2/3(e) - 1/3(e)$ = $(+)1(e)$	A1

60.

7(a)	down charge = $-1/3(e)$ and charm charge = $(+)2/3(e)$	B1
	all antiquarks have opposite sign and same (non-zero) magnitude of charge as the corresponding quarks	B1
7(b)(i)	udd or cdd	B1
7(b)(ii)	$u\bar{d}$ or $c\bar{d}$	B1
7(c)(i)	lepton(s)	B1
7(c)(ii)	positron / neutrino / antineutrino	B1

61.

7(a)	a (very) small proportion of (alpha) particles are deflected (back) through large angles / angles greater than 90°	B1
	a large proportion of (alpha) particles pass straight through / deflected by small angles	B1
7(b)	electromagnetic wave / electromagnetic radiation	A1
7(c)	neutrino classified as a lepton only <u>and</u> positron classified as a lepton only	B1
	neutron classified as a baryon and a hadron and not as a lepton	B1
7(d)(i)	1.32 u	A1
7(d)(ii)	working states or implies 3 quarks and each quark has a charge of $(+) \frac{2}{3}(e)$	B1
	any combination of 3 quarks comprised of one or more of up / charm / top	B1

62.

7(a)	only baryon and nucleon underlined	B1
7(b)	number of α -particles = 6	B1
	number of β^- particles = 4	B1
7(c)	charm antiquark (charge) = $-2/3(e)$ or $-2/3(e) + q = -1(e)$	B1
	(other quark $q = -1/3(e)$ so) down / strange / bottom	B1

63.

8(a)(i)	23: nucleon number or number of neutrons and protons	B1
	12: proton number or number of protons	B1
8(a)(ii)	both numbers correct for X: $^{23}_{11}\text{X}$	B1
	both numbers correct for beta-plus: $^0_1\beta^+$	B1
8(a)(iii)	lepton(s)	A1
8(b)	(the emitted particles) have a single (kinetic) energy or beta particles / decay have a (continuous) range of (kinetic) energies	B1